

SRM VEC
DEPARTMENT OF AIDS
Semester End Examination
(Regulations 2023)
AD3445 – AI1

Branch: B.Tech | Date: 01.02.2026 | Max. Marks: 100 | Duration: 3 Hrs

Answer ALL Questions
(100 Marks)

PART A — (10 × 2 = 20 marks)

1. Define a Rational Agent and specify the four factors that depend on rationality. [CO1] [BTL1-Remember] (2)
2. Differentiate between Informed and Uninformed search strategies with one example for each. [CO2] [BTL2-Understand] (2)
3. What are the requirements for a heuristic function to be 'admissible' in A* search? [CO2] [BTL1-Remember] (2)
4. State the purpose of Alpha-Beta pruning in game playing. [CO2] [BTL2-Understand] (2)
5. Convert the following sentence into First-Order Logic (FOL): 'Every student who takes AI likes logic.' [CO3] [BTL3-Apply] (2)
6. Define the term 'Skolemization' in the context of converting FOL to Clausal Form. [CO3] [BTL1-Remember] (2)
7. State the Bayes' Rule and explain each term in the formula. [CO4] [BTL1-Remember] (2)
8. What is a Markov Blanket in a Bayesian Network? [CO4] [BTL2-Understand] (2)
9. List the components of a STRIPS planning problem. [CO5] [BTL1-Remember] (2)
10. Distinguish between Supervised and Unsupervised Learning. [CO5] [BTL2-Understand] (2)

PART B — (5 × 16 = 80 marks)

11(a) (i) Explain the different types of Intelligent Agents with neat diagrams. (ii) Formulate the PEAS description and environment properties for an 'Automated Medical Diagnosis System'.

OR

(b) (i) Explain the A* Search algorithm. Prove that A* is optimal if the heuristic is admissible. (ii) Compare Breadth-First Search and Depth-First Search based on Time Complexity, Space Complexity, Completeness, and Optimality.

12(a) Consider a game tree where the root is a MAX node. Apply the Minimax algorithm with Alpha-Beta pruning to find the optimal move. Assume the leaf nodes (left to right) are: 3, 12, 8, 2, 4, 6, 14, 5, 2. Show the pruned branches and the updated alpha/beta values at each step.

OR

(b) (i) Define Constraint Satisfaction Problem (CSP). Explain the Backtracking search for CSP. (ii) Solve the 4-Queens problem using backtracking search. Show the state space tree.

13(a) (i) Explain the process of Unification in First-Order Logic with a suitable example. (ii) Use Resolution to prove that 'Marcus is dead' given: 1. Marcus is a man. 2. Marcus is a Pompeian. 3. All Pompeians died in the eruption in 79 AD. 4. It is now 2024 AD. 5. All men die after 150 years.

OR

(b) (i) Compare and contrast Forward Chaining and Backward Chaining with examples. (ii) Explain the components of a Frame-based knowledge representation system. Represent the concept 'University' using Frames.

14(a) Given the following Bayesian Network: Weather (W) affects whether a person carries an Umbrella (U). Whether the person carries an Umbrella and whether it is Raining (R) affects whether the person gets Wet (T). (i) Draw the Bayesian Network. (ii) Explain the concept of Conditional Independence in this network. (iii) Calculate $P(T | W, R)$ using the chain rule and joint probability distribution.

OR

(b) (i) Explain the Variable Elimination algorithm for exact inference in Bayesian Networks. (ii) Discuss the Hidden Markov Model (HMM) and its application in speech recognition.

15(a) (i) Explain the working of the ID3 Decision Tree induction algorithm using Information Gain and Entropy. (ii) Describe the components of a Planning Graph and how it is used to estimate heuristic costs.

OR

(b) (i) Explain the Partial-Order Planning (POP) algorithm with an example. How does it differ from Total-Order Planning? (ii) Briefly explain the concept of Reinforcement Learning with the help of the Agent-Environment interface.